# 👌 XAI Lecture 22

## Compiled Transformers as a Laboratory for Interpretability

FAMAF - UNC

First Semester 2026

# Disclaimer ---

**This course is based on**

**Explainable Artificial Intelligence**

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

# Tracr: Compiled Transformers as a Laboratory for Interpretability

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**Abstract**

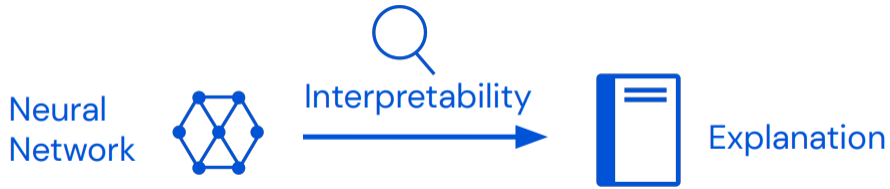
We show how to "compile" human-readable programs into standard decoder-only transformer models. Our compiler, `Tracr`, generates models with known structure. This structure can be used to design experiments. For example, we use it to study "superposition" in transformers that execute multi-step al-

(Lindner et al. 2023)

# Motivations - Tracr ---

- **Controlled experiments:** Creates simple, controlled Transformer models from RASP programs.
- **Ground truth availability:** Provides known internal mechanisms to evaluate interpretability methods.
- **Bridge algorithms $\leftrightarrow$ Transformers:** Shows how algorithms can be implemented inside attention-based models.
- **Reduction of confounding factors:** Avoids training noise and unclear learned objectives.
- **Educational and research utility:** Works as a laboratory for mechanistic interpretability.

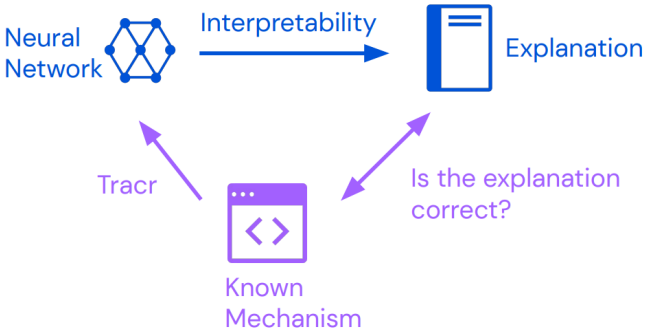
# The Core Problem: No Ground Truth ---




**Interpretability bottleneck**

For a trained neural network, there is usually no independent ground-truth mechanism telling us whether an explanation is correct.

# What If We Had Ground Truth? ---




If the network is compiled from a **known mechanism**, then an explanation can be checked against the program that generated the weights.

# Introducing Tracr ---

Known Mechanism ⟶ Neural Network

Tracr is a compiler from **RASP programs** into **decoder-only transformer weights**.

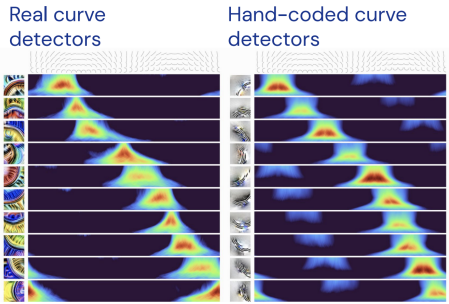
(Lindner et al. 2023)

# Plan for Today ---

1. Building a **compiler** for transformer models

2. Studying **superposition** in compiled models

# Why Not Hand-Code Weights? ---




(Cammarata et al. 2021)

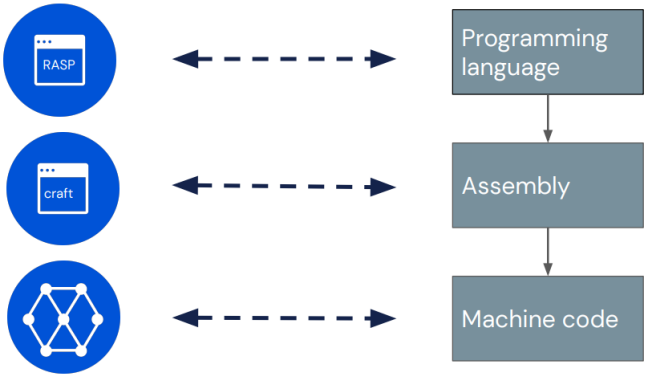
Hand-coding weights is a strong form of understanding:

- We specify the circuit directly
- We know what each component should do
- We can compare learned behavior against the construction

**Problem**

This does not scale. It is like programming a model in bytecode.

# Part 1: Building the Compiler ---




Tracr works analogously to how we would translate a programming language into executable code.

# Three-Step Translation ---


**Human readable code in domain-specific language**


**Basis independent representation of vector spaces and transformers**

**Neural network weights**

## Design principle

Keep the high-level algorithm explicit as long as possible, then lower it into a standard transformer implementation.

# RASP: A Language for Transformer Computations ---

**Definition**

**RASP** = "Restricted Access Sequence Processing Language"


- Two variable types: **sequence operations** and **selectors**
- Two instruction types: **elementwise** and **select-aggregate**
- S-ops roughly correspond to transformer **residual stream state**

**Example sequence**

bos x a c x

**Primitive s-ops**

`tokens("hello")` $= [h, e, l, l, o]$

`indices("hello")` $= [0, 1, 2, 3, 4]$

# RASP Instructions Map to Transformer Blocks ---

## Elementwise operations

- Apply a function independently at each sequence position
- Example: `(3 * indices)("hello")` $= [0, 3, 6, 9, 12]$
- Roughly correspond to transformer **MLP layers**

## Select-aggregate operations

- Move information between token positions
- A selector evaluates to an $N \times N$ binary matrix
- `select` builds the selector; `aggregate` reads selected values
- Roughly correspond to transformer **attention**

## Compiler intuition

RASP keeps the algorithm readable, while Tracr lowers these operations into transformer components.

# RASP Example: Count Previous x Tokens ---

## RASP code

```
is_x = (tokens == "x")
prevs = select(indices, indices, <=)
frac_prev = aggregate(prevs, is_x)
```

```
seq := ["a", "x", "b", "x", "c"]
tokens(seq) = ["a", "x", "b", "x", "c"]
indices(seq) = [0, 1, 2, 3, 4]

is_x(seq) = [0, 1, 0, 1, 0]
prevs(seq) = [[1, 0, 0, 0, 0],
              [1, 1, 0, 0, 0],
              [1, 1, 1, 0, 0],
              [1, 1, 1, 1, 0],
              [1, 1, 1, 1, 1]]
frac_prev(seq) = [0, 1/2, 1/3, 2/4, 2/5]
```

## In plain English

For each position, ask: *among all previous positions including this one, what fraction contained the token x?*

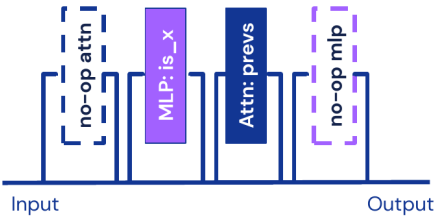
# From RASP to transformer weights ---

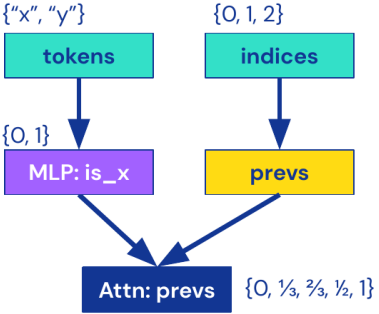
## Compilation process (6 steps)

1. Construct computational graph.
2. Infer s-op input/output values.
3. Translate s-ops into model blocks.
4. Assign components to layers.
5. Assemble the craft transformer.
6. Generate weight matrices.

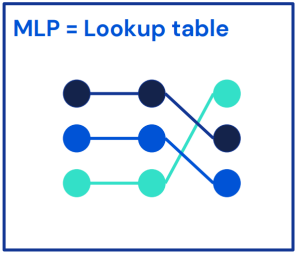
## RASP code

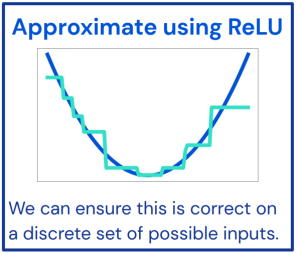
```
is_x = (tokens == "x")
prevs = select(indices, indices, <=)
frac_prev = aggregate(prevs, is_x)
```



# Implementing MLP Layers ---




## Categorical variables

An MLP can implement a lookup table over finitely many categories.

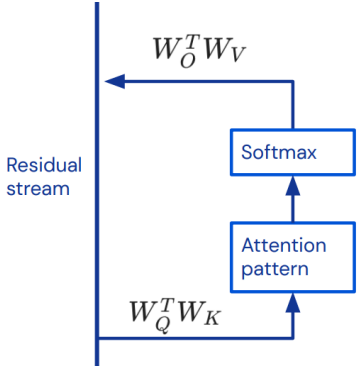
## Numerical variables

A ReLU network can approximate the desired function exactly on the discrete input set used by the program.

# Implementing Attention Heads ---

## Operation mapping

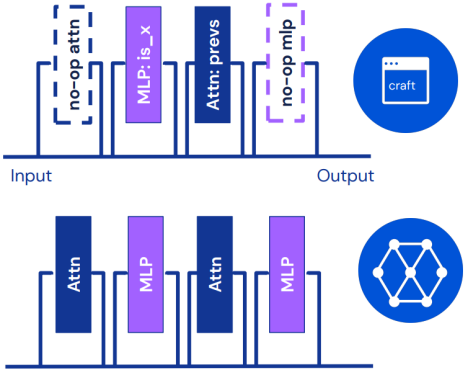
| RASP operation | Transformer component |
|---|---|
| `select(indices, indices, <=)` | query key scores $W_Q^T W_K$ |
| `aggregate(prevs, is_x)` | value output map $W_O^T W_V$ |




**In plain English**

Attention is used as programmable routing: decide *which positions to read from*, then aggregate their values.

# craft Models Map to Standard Transformers ---




## Key idea

craft separates the abstract vector-space computation from implementation details, then maps it into a GPT-like transformer architecture.

# What Can Tracr Compile? ---

**Examples**

- Count tokens and build histograms
- Detect all occurrences of a pattern
- Sort a sequence
- Check balanced parentheses
- ...

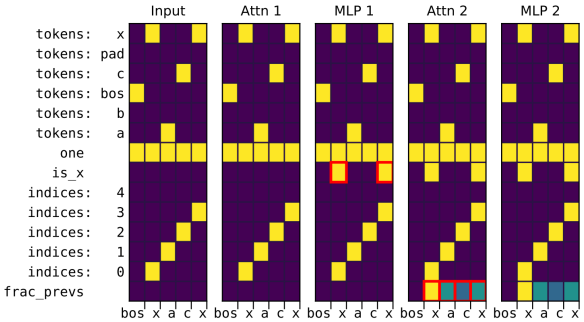
**Limitations**

- Binary attention patterns
- Algorithmic, not probabilistic, tasks
- Programs close to transformer structure
- Large and inefficient compiled models

# A Compiled Residual Stream ---

## RASP program

```
is_x = (tokens == "x")
prevs = select(indices, indices, <=)
frac_prev = aggregate(prevs, is_x)
```




## Key idea

The residual stream exposes the compiled variables: first the MLP computes `is_x`, then attention computes the fraction over previous positions.

# What Does Ground Truth Buy Us? ---

With a compiled model, we know:

- Which variables should exist
- Which layer computes each variable
- Which attention head implements each selector
- Which features are necessary for the final output

**Evaluation idea**

Run an interpretability method and ask whether it recovers the same components the compiler inserted.

**Caveat**

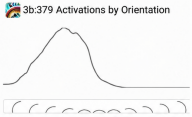
This only tests interpretability on synthetic algorithmic models, not arbitrary trained LLMs.

2. Studying **superposition** in compiled models

# The Superposition Hypothesis ---

## Observation 1

Some neurons correspond to clean, interpretable features.




Cammarata, et al., "Thread: Circuits", Distill, 2020.

## Observation 2

Other neurons appear to represent multiple features.

Simple Optimization Dataset examples

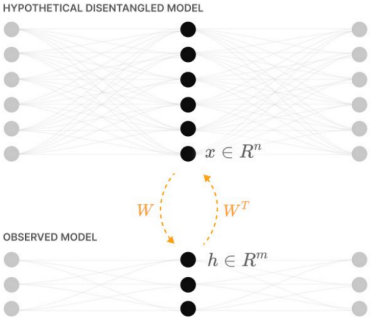
Olah, et al., "Feature Visualization", Distill, 2017.

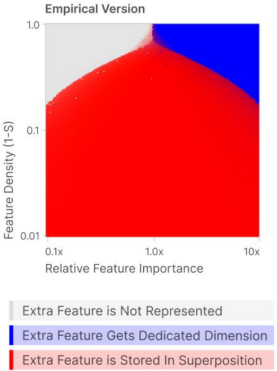
## Observation 3

A linear representation can store more features than dimensions when features are sparse.


Elhage, et al., "Toy Models of Superposition", Transformer Circuits Thread, 2022.

# Toy Models Show When Superposition Appears ---




$$h = Wx$$
$$x' = \text{ReLU}(W^T h + b)$$
$$x' = \text{ReLU}(W^T W x + b)$$

$$L = \sum_x \sum_i I_i (x_i - x'_i)^2$$

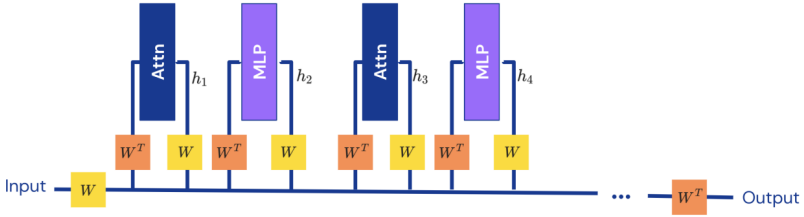
(Elhage et al., "Toy Models of Superposition", Transformer Circuits Thread, 2022.)

# Why Compress Tracr Models? ---

| In toy models we see superposition if | | In Tracr models |
|---|---|---|
| 1. Features are **sparse** | $\Longrightarrow$ | 1. Features are **sparse** |
| 2. Some features are more **important** than others | $\Longrightarrow$ | 2. Some features are more **important** for the computation |
| 3. The model has to use **fewer dimensions than features** | $\Longrightarrow$ | 3. Can we **compress** the model to use **fewer dimensions**? |

Tracr models already have sparse, meaningful variables. If we force their residual stream to use fewer dimensions, we can study superposition with a known feature basis.

# Linear Compression Objective ---




We train $W \in \mathbb{R}^{D \times d}$ to minimize:

$$\mathcal{L}_{\text{total}}(W) = \mathbb{E}_x \left[ \mathcal{L}_{\text{out}}(W, x) + \mathcal{L}_{\text{layer}}(W, x) \right]$$

$$\mathcal{L}_{\text{out}}(W, x) = \text{loss}\left( f(x), \hat{f}_W(x) \right)$$

**minimize output loss**

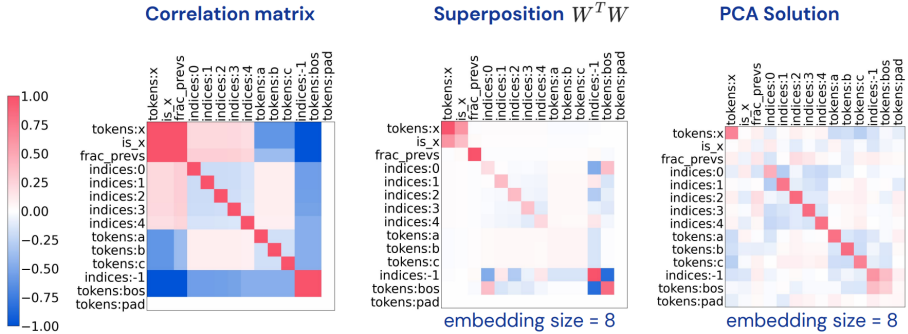
$$\mathcal{L}_{\text{layer}}(W, x) = \sum_{\text{layer } i} \left( h_i(x) - \hat{h}_{W,i}(x) \right)^2$$

**implement the same computation**

## In plain English

Train only a compression map $W$: the compressed model should preserve the final output and keep each intermediate layer close to the original compiled computation.

# What Changes After Compression? ---

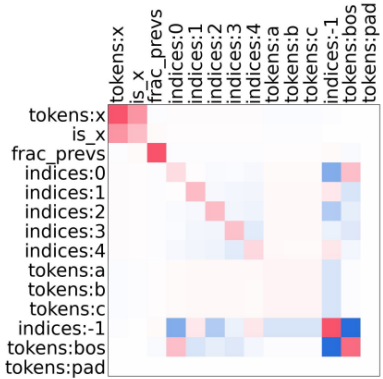



The learned $W^TW$ structure is qualitatively different from PCA: less important and more independent features are packed into shared directions.

# Which Features Enter Superposition? ---

The lecture highlights three drivers:

- **Importance:** how much the computation needs the feature
- **Density:** how often the feature is active
- **Linear independence:** how easily it can share directions with others



# Discussion Questions ---

- **Can Tracr create evaluation benchmarks** for interpretability tools?
- **Can we reverse superposition** in Tracr models?
  *Sparse coding, dictionary learning*
- **Can we manually replace model components** we think we understand?

# Summary ---

- Tracr compiles RASP programs into transformer weights
- The compiled model has a known mechanism
- This gives interpretability methods an answer key
- Compression induces superposition in a controlled feature basis
- The main value is experimental control, not realism

**Resources**

https://github.com/google-deepmind/tracr

https://arxiv.org/abs/2301.05062

# Thank You!

# References ---


Lindner, David, János Kramár, Sebastian Farquhar, Matthew Rahtz, Thomas McGrath, and Vladimir Mikulik. 2023. "Tracr: Compiled Transformers as a Laboratory for Interpretability." arXiv:2301.05062.

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Olah, Chris, Alexander Mordvintsev, and Ludwig Schubert. 2017. "Feature Visualization." Distill.

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